

Now you're entering the **implementation phase**, so the next artifact is not another business document.

The next document is:

SQL Schema Design Document (SSDD)

ScrapFlow AI v1.0

This converts your DDD into actual MySQL implementation.

Purpose

Define:

- CREATE TABLE statements
 - Foreign Keys
 - Constraints
 - Indexes
 - Default Values
 - Triggers (future)
-

Database Creation

```
CREATE DATABASE scrapflow_db;
```

```
USE scrapflow_db;
```

Table Creation Order

Very important.

1. users
2. buyers
3. batches
4. materials

5. inventory

6. expenses

7. sales

Because foreign keys depend on parent tables.

Users Table

Need specification for:

Primary Key

Unique Email

Role Enum

Created Timestamp

Buyers Table

Need specification for:

Buyer Details

Phone Validation

Address

Notes

Batches Table

Need specification for:

Foreign Key → users

Status Enum:

Collected

Sorting

Sorted

Partially Sold

Sold

Materials Table

Need specification for:

Foreign Key → batches

Material Type Enum

Weight Precision

Inventory Table

Need specification for:

Material Type

Available Weight

Average Cost

Estimated Value

Expenses Table

Need specification for:

Foreign Key → batches

Expense Category

Amount

Date

Sales Table

Need specification for:

Foreign Key → buyers

Foreign Key → materials

Quantity

Price Per Kg

Total Amount

Status

Required Constraints

Users

UNIQUE(email)

Materials

weight > 0

Expenses

amount > 0

Sales

quantity > 0

price_per_kg > 0

Required Indexes

For performance:

users(email)

batches(collection_date)

materials(material_type)

buyers(phone)

sales(sale_date)

Seed Data Design

For development:

Material Types

Copper

Aluminum

Steel

Iron

Brass

Plastic

Other

Expense Types

Fuel

Transport

Maintenance

Miscellaneous

Future SQL Objects

Version 2:

Views

Stored Procedures

Triggers

Example:

profit_summary_view

inventory_summary_view

After SSDD

The next step is no longer documentation.

The sequence becomes:

09 SQL Schema Design (Current)

10 MySQL Implementation

11 FastAPI Project Setup

12 SQLAlchemy Models

13 Authentication Module

14 Batch APIs

15 Inventory APIs

16 Sales APIs

17 Dashboard APIs

18 Frontend

What I would do now

Do **not** create more planning documents.

Your planning stack is already stronger than many startup teams.

Your next real deliverable should be:

schema.sql

containing all actual MySQL CREATE TABLE, ALTER TABLE, indexes, and foreign key statements.

After that, start:

FastAPI Project Setup

because that's where the project begins to become a real product rather than documentation.

